Grishma Khatra

Thunder Bay, ON — +1 (437) 655-0273 — grishmakhatra93@gmail.com
linkedin.com/in/grishma-khatra — github.com/Grishmakhatra932

Summary

Machine Learning Engineer and Master's **Computer Science** student with hands-on experience building **LLM-based AI systems**, **Multi-Agent Architectures**, and **Recommendation Systems**. Skilled in **Python**, **Natural Language Processing (NLP)**, **LangChain**, **Large Language Models (LLMs)**, and **Model Deployment** using **Docker** and **AWS**. Passionate about building **Scalable AI Solutions** and **Production-Ready Machine Learning Systems**.

Technical Skills

- **Programming:** Python, SQL, C#
- **Machine Learning & AI:** Machine Learning, Deep Learning, NLP, LLMs, Retrieval-Augmented Generation (RAG), Prompt Engineering, AI Agents, Recommendation Systems
- **Frameworks & Libraries:** LangChain, LangGraph, Hugging Face Transformers, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
- **Backend & APIs:** FastAPI, REST APIs, API Integration, Model Inference APIs
- **Cloud, DevOps & MLOps:** AWS (ECR, ECS), Docker, Jenkins, CI/CD Pipelines, SonarQube, Git, GitHub
- **Monitoring & Data Tools:** Grafana, Prometheus, Power BI, Vector Embeddings, Semantic Search

Projects

Multi-AI Agent System **GitHub**

- Built an end-to-end **LLM-powered Multi-AI Agent application** using **Python and Streamlit** enabling users to create custom AI agents and interact through a dynamic UI.
- Integrated **Groq API** for large language model inference and **Tavily API** for optional real-time web search capabilities.
- Designed a **prompt-based AI agent workflow** allowing users to define agent roles, select LLM models, and generate contextual responses.
- Implemented complete **CI/CD pipeline** using **GitHub, Jenkins, and SonarQube** for automated build, testing, and quality analysis.
- Containerized application using **Docker** and deployed on **AWS ECS** with container images stored in **AWS ECR**.

AI Anime Recommender System with Grafana **GitHub**

- Developed an **AI-powered recommendation system** using **vector embeddings**, **semantic similarity**, and personalized ranking for content recommendation.
- Built a **Python REST API** for query processing, recommendation retrieval, and scalable inference workflow.
- Improved recommendation quality using **embedding-based vector search** over traditional keyword-based retrieval methods.
- Integrated **Grafana** and **Prometheus** for API monitoring, latency tracking, and system performance visualization.

Experience

Hubbroker IT Solutions Private Limited **Feb 2023 – Jul 2025**
Technical Solution Engineer *Ahmedabad, India*

- Developed and supported **ERP** and **Electronic Data Interchange (EDI)** solutions for B2B clients using **Dynamics 365, Shopify, Rackbeat, and e-conomic**.
- Designed, tested, and validated **REST APIs** for system integration, business process automation, and data exchange across ERP and webshop platforms.
- Created QA documentation and test cases, improving software quality, workflow accuracy, and issue resolution efficiency.
- Led **PDF-to-XML** and **EDI document transformation** workflows using **HUBXML, OIOUBL, BISV3, and EDIFACT**, ensuring standards compliance and reduced manual processing.

Education

Lakehead University **Sep 2025 – Aug 2027**
Master of Computer Science *Thunder Bay, Ontario, Canada*

Parul Institute of Engineering and Technology **2019 – 2023**
B.Tech in Computer Science and Engineering (PPI: 7.40/10) *Vadodara, Gujarat, India*